

Proposed work: Analysis of the field GNSS data collected in the OAUP.

Objectives: Analysis of the signals from the field survey observations with RTKPLOT.

1. In order to give answers to the exercises proposed in this sheet, identify, in the set of RINEX files provided, those corresponding to your points of the field survey. Always confirm the 1st epoch that appears in the RINEX file with the start time indicated on the field sheet.
2. Analyze the GNSS signals observed during fieldwork at the points assigned to you for the receivers listed below. With the RTKPLOT application, create and save the graphs for satellite visibility (Sat Vis), visibility diagram (Skyplot), geometry and number of satellites (DOP/NSat), signal strength as a function of time (SNR/MP/EL) and as a function of elevation (SNR/MP-EL). The images of the graphs must be included in the final report.
 - 2.1. Septentrio PolaRx5 geodetic receiver.
 - 2.2. Xiaomi Mi8 Smartphone.
 - 2.3. EMLID Reach M2 receiver (low cost).
 - 2.4. Septentrio PolaRx5 geodetic receiver (kinematic).
3. Fill in the table below according to the parameters analyzed in the graphs obtained in the previous paragraphs. Indicate the most relevant differences between the different receivers.

Receiver GNSS	Point	Constellations	# Frequencies	# Minimum Satellites	# Maximum satellites	Highest % Satellites	Average PDOP
<i>Septentrio PolaRx5</i>	PL3	G, R, E, C, S	5	14	40	39(45.5%)	0.8
<i>Xiaomi Mi8</i>	PL2	G, R, E, C	2	21	27	24: 39.5%	—
<i>EMLID Reach M2</i>	M3	G, R, E, C, S	2	15	41	40: 39.2%	0.7
<i>Septentrio PolaRx5</i>	RED	G, R, E, C, S	5	20	40	38 (36.2%)	0.8

The Septentrio PolaRx5 shows the best overall performance, with the most frequencies, high satellite counts, and stable low PDOP (~0.8).

The EMLID Reach M2 also performs very well, achieving the highest number of satellites (41) and the lowest PDOP (0.7), despite using fewer frequencies.

The Xiaomi Mi8 performs the worst, with fewer tracked satellites and limited frequency support, resulting in lower positioning reliability.

Overall, professional GNSS receivers clearly outperform the smartphone.